

Exercises sheet 2: Inclusion-Exclusion principle

Exercise 1. 51 numbers are chosen from the integers between 1 and 100 inclusively. Prove that 2 of the chosen integers are consecutive.

Exercise 2. There are 5 points in a square of side length 2 units. Prove that there exists 2 of them having a distance at most $\sqrt{2}$.

Exercise 3. Show that for any set of 10 points chosen within a square whose sides are of length 3 units, there are two points in the set whose distance is at most $\sqrt{2}$.

Exercise 4. Sixty five points are given inside a square of side length one unit. Prove that there are three of them that span a triangle of area at most $1/32$.

Exercise 5. Let $S = \{1, \dots, 20\}$, if we pick eleven numbers, we are guaranteed that the sum of two picked numbers is 21.

Exercise 6. Show that among any group of n people, where $n \geq 2$, there are at least two people who know exactly the same number of people in the group.

Exercise 7. Show that any set of 7 distinct integers includes 2 integers x and y such that either $x + y$ or $x - y$ is divisible by 10.

Exercise 8. The total number of games played by a team in a 15-day season was 20. The rules required the team to play at least 1 game daily. Show that there was a period of consecutive days during which exactly 9 games were played.

Exercise 9. Let us arbitrarily select $n + 1$ distinct integers from the set $[2n] = \{1, 2, \dots, 2n\}$. Then there is at least one pair of selected integers whose sum is $2n + 1$, and there is at least one pair of selected integers whose difference is n .

Exercise 10. Show that any set of n integers has a subset such that the sum of integers in the subset is divisible by n .

Exercise 11. Let A be $n \times n$ matrix with 0 and 1 entries only. Let us assume that $n \geq 2$, and that at least $2n$ entries are equal to 1. Prove that A contains two entries equal to 1 so that one of them is strictly above and strictly at the right of the other.

Exercise 12. Let $A = \{a_1, a_2, a_3, a_4, a_5\}$ be a set of 5 integers. Show that for any permutation $a_{i_1} a_{i_2} a_{i_3} a_{i_4} a_{i_5}$ of A , the product $(a_{i_1} - a_1)(a_{i_2} - a_2) \cdots (a_{i_5} - a_5)$ is always even.

Exercise 13. *There are 15 subjects in a school, every student takes 4 subjects out of them. Given that the school have 18 students, prove that 2 of the students have two common subjects.*

Exercise 14. *Let $S \subseteq \{1, 2, \dots, 2n\}$ such that $|S| = n + 1$, where $n \in \mathbb{N}$. Show that there exist $a, b \in S$, with $a \neq b$ such that $a|b$.*

Exercise 15. *41 rooks are placed on a 10×10 -chessboard. Prove that there are five rooks among these 41 that do not attack each other (i.e., no two of them are in the same horizontal or vertical line).*

Exercise 16. *Five lattice points are chosen on an $n \times n$ square lattice. Line segments are drawn between every pair of these points. Prove that one of the midpoints of these line segments is also a lattice point.*

Exercise 17. *Every point in a plane is either red, green, or blue. Prove that there exists a rectangle in the plane such that all of its vertices are the same color.*

Exercise 18. *What is the number of integral solutions of the equation*

$$x_1 + x_2 + x_3 + x_4 = 18,$$

that satisfy

$$1 \leq x_1 \leq 5, \quad -2 \leq x_2 \leq 4, \quad 0 \leq x_3 \leq 5, \quad 3 \leq x_4 \leq 9.$$

Exercise 19. *Let $S(n, k)$, $n, k \in \mathbb{N}$, denote the number of surjective mappings from $\mathbb{N}_n$ to $\mathbb{N}_k$. Show that*

$$S(n, k) = \sum_{i=0}^k (-1)^i \binom{k}{i} (k-i)^n.$$

Exercise 20. *How many permutations of the 26 letters do not contain any of the following sequences: MATH, EXA, DISCR, NHSM.*

Exercise 21. *Let $D(n)$ be the number of derangements for n different objects, using the inclusion-exclusion principle, show that*

$$D(n) = n! \sum_{k=0}^n \frac{(-1)^k}{k!}.$$

Exercise 22. *How many permutations of the word "discretemaths" does not contain consecutive pairs?*

Exercise 23. *Let $k, n, r \in \mathbb{N}$. Find the number of integer solutions to the equation*

$$x_1 + x_2 + \dots + x_n = r,$$

such that $0 \leq x_i \leq k$, for each $i = 1, 2, \dots, n$.

Exercise 24.

- (1) Let B be a subset of A with $|A| = n$ and $|B| = m$. Find the number of r -element subsets of A which contain B as a subset, where $m \leq r \leq n$.
- (2) Show that for $m, r, n \in \mathbb{N}$ with $m \leq r \leq n$,

$$\binom{n-m}{n-r} = \sum_{i=0}^m (-1)^i \binom{m}{i} \binom{n-i}{r}$$

Exercise 25. Show that

$$\binom{n}{k} = \sum_{m=0}^{\lfloor k/2 \rfloor} (-1)^m \binom{n}{m} \binom{n+k-2m-1}{n-1}$$

by a suitable combinatorial interpretation of the binomial coefficients and application of the inclusion-exclusion principle.

Exercise 26. Let H_n be the number of ways of seating n married couples around a circular table so that no man is next to his wife. Show that

$$H_n = \sum_{k=0}^n (-1)^k \binom{n}{k} 2^k (2n-k-1)!$$

Exercise 27. Let W_n be the number of ways of seating n married couples at a straight table so that no woman is next to her husband. Show that

$$W_n = \sum_{k=0}^n (-1)^k \binom{n}{k} 2^k (2n-k)!$$

Exercise 28. Let $T_{r,s}(n, k)$ be the number of k -combinations of n with repetition and the restriction that each element appears at least r and at most s times. Show that

$$T_{r,s}(n, k) = \sum_{j=0}^n (-1)^j \binom{n}{j} \binom{n+k-rn-j(s-r+1)-1}{n-1}.$$

Exercise 29. Consider $2n$ elements that belong to n different kinds, with two like elements of each kind, and let V_n be the number of permutations of these elements in which no two like elements are consecutive. Show that

$$V_n = 2^{-n} \sum_{k=0}^n (-1)^k \binom{n}{k} 2^k (2n-k)!$$